

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 10: Respiration in Organisms

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	10 — Respiration in Organisms
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: equating breathing with cellular respiration, assuming exhaled air has zero oxygen or fresh air zero carbon dioxide, believing plants stop respiring by day or do not respire at all, treating anaerobic respiration as releasing as much energy as aerobic, reversing diaphragm and rib movements or the pressure logic of inhalation/exhalation, swapping the breathing organs of different organisms (gills/spiracles/tracheae/skin/stomata/lenticels), and reading breathing-rate or gas-volume data backwards. Do not write on this paper — mark answers on your answer sheet.

1. A teacher writes on the board: 'In our cells, food is broken down with oxygen, and energy is set free.' A student asks why the body cannot simply absorb energy ready-made from the food it eats. The best Class-7 answer is that:

- (A) the energy is locked inside the chemical bonds of glucose and must be released by breaking the glucose down during respiration
- (B) food contains no energy until oxygen is added to it in the lungs
- (C) energy can only be made fresh in the cells, never obtained from food
- (D) the stomach destroys all the energy in food, so cells must remake it

2. Two cells use the same amount of glucose. Cell M is supplied with plenty of oxygen; cell N has almost none. Which cell releases MORE energy from the glucose, and why?

- (A) cell N, because working without oxygen forces it to squeeze out more energy
- (B) both release exactly the same energy, since they start with equal glucose

- (C) cell N, because it produces alcohol which stores extra energy
- (D) cell M, because with oxygen the glucose is broken down completely to carbon dioxide and water, releasing the most energy

3. A student writes: 'When I run upstairs my chest heaves faster, which proves my lungs are releasing more energy.' The single best correction is:

- (A) the faster chest movements are breathing, which only exchanges air; the extra energy is released by respiration inside the muscle cells
- (B) the lungs do release the extra energy, but only while you are moving
- (C) faster breathing actually slows the release of energy in the cells
- (D) breathing and energy release are unconnected, so the heaving means nothing

4. Trace one oxygen molecule from the moment it enters the nose until it reaches the blood. The correct ordered path is:

- (A) nostrils → nasal cavity → trachea → bronchi → alveoli → blood
- (B) nostrils → trachea → nasal cavity → alveoli → bronchi → blood
- (C) nasal cavity → nostrils → bronchi → alveoli → trachea → blood
- (D) nostrils → nasal cavity → bronchi → trachea → alveoli → blood

5. A child argues, 'A seed is not breathing, so it must be dead and cannot respire.' Which observation BEST refutes this and shows a dormant-looking soaked seed still respire?

- (A) the seeds feel cool and dry, proving no chemical activity inside them
- (B) soaked germinating seeds sealed with limewater turn the limewater milky, showing they release carbon dioxide
- (C) a candle placed beside the seeds in fresh air keeps burning brightly
- (D) the seeds float on water, which only living things can do

6. A runner's calf cramps after a fast 400 m dash, then eases after a few minutes of rest. Which sequence correctly explains the whole episode?

- (A) muscles made alcohol while running, and resting evaporated the alcohol away
- (B) oxygen supply lagged behind, so muscles respired anaerobically and built up lactic acid; resting let oxygen reach the muscle and break the lactic acid down
- (C) resting made the muscle produce extra lactic acid, which loosened the cramp
- (D) the cramp came from too much oxygen, and rest lowered the oxygen level

7. Warm sugar dough with yeast is divided into two equal lumps: lump P is left open in warm air; lump Q is sealed tightly in a jar with no extra air. Both rise. The gas that raises BOTH lumps and the process that makes it are:

- (A) oxygen, from the yeast photosynthesising the sugar
- (B) carbon dioxide, from the yeast respiring sugar anaerobically
- (C) water vapour, from the warmth boiling the dough
- (D) carbon dioxide, from the yeast photosynthesising in the light

8. After hard exercise, why does an athlete keep panting (breathing deeply and quickly) for several minutes even after standing still?

- (A) the lungs are stuck open from the exercise and cannot slow down yet
- (B) the muscles keep making alcohol that the fast breathing blows away
- (C) the body still needs extra oxygen to break down the lactic acid that built up during the effort
- (D) the body must keep breathing fast to go on making more lactic acid

9. A table records breathing rates for one boy: asleep 13, sitting 16, walking quickly 24, sprinting 40 (breaths per minute). The reading that BEST supports the table is:

- (A) breathing rate is highest when the body is most at rest
- (B) breathing rate falls steadily as activity increases
- (C) as the muscles work harder, the body breathes faster to supply more oxygen and remove more carbon dioxide
- (D) the body breathes fastest while asleep because it stores oxygen for the day

10. A model lung uses a bottle with a stretchy rubber sheet across the open bottom and a balloon hanging inside. Pushing the sheet UP into the bottle makes the balloon deflate. In a real body, this models:

- (A) inhalation, where the diaphragm flattens and the chest expands to draw air in
- (B) exhalation, where the diaphragm relaxes and domes up, shrinking the chest and pushing air out
- (C) the heart pumping blood through the chest, not breathing at all
- (D) inhalation, because pushing the sheet up enlarges the space for the balloon

11. Two jars of clear limewater are set up. Jar X is bubbled with air a student breathes OUT; jar Y is bubbled for the same time with ordinary room air pumped in. X turns milky much faster than Y, but Y also turns slightly milky after a long time. The fairest conclusion is:

- (A) both exhaled air and room air contain carbon dioxide, but exhaled air contains much more
- (B) only exhaled air contains carbon dioxide; room air contains none at all
- (C) room air contains more carbon dioxide than exhaled air, since it turned milky too
- (D) neither sample contains carbon dioxide; the milkiness is caused by water vapour

12. Inhaled air is about 21% oxygen and 0.04% carbon dioxide; exhaled air is about 16% oxygen and 4% carbon dioxide. Which conclusion is correct?

- (A) the body uses up some oxygen and adds a much larger share of carbon dioxide, but plenty of oxygen still remains in the exhaled air
- (B) the body removes all the oxygen, so exhaled air contains none
- (C) the body adds oxygen and removes carbon dioxide as the air passes through
- (D) inhaled and exhaled air have the same gas amounts, so respiration changes nothing

13. Two same-age students do identical step-ups. Afterwards student A breathes 22/min and student B breathes 34/min, and A is the trained sports player. The most reasonable reading is:

- (A) A's fitter body delivers oxygen more efficiently, so A needs fewer breaths to do the same work
- (B) B is fitter, because a higher breathing rate always means stronger lungs
- (C) A's cells have stopped respiring, which is why A breathes less
- (D) A breathes less because trained people no longer need oxygen during exercise

14. Grape juice with baker's yeast is left in a warm covered flask. After two days it smells of alcohol, fizzes, and a balloon over the neck has inflated. Which single explanation accounts for ALL three observations?

- (A) the yeast photosynthesised the sugar, giving off oxygen that filled the balloon
- (B) the warm juice boiled, the steam smelled sweet, and water vapour filled the balloon
- (C) the yeast turned the alcohol back into sugar, releasing oxygen
- (D) the yeast respired the sugar without enough oxygen, producing alcohol and carbon dioxide gas

15. A man breathing 15 times per minute keeps that rate steady. About how many breaths does he take in three quarters of an hour (45 minutes)?

- (A) about 675
- (B) about 60
- (C) about 6750
- (D) about 450

16. During a normal inhalation, what happens to the diaphragm and ribs, and what is the effect on the space inside the chest?

- (A) the diaphragm domes up while the ribs move down, so the chest space gets smaller
- (B) the diaphragm flattens down while the ribs move down and in, so the chest space gets smaller
- (C) the diaphragm and ribs both stay still while the lungs pull the air in by themselves
- (D) the diaphragm flattens and moves down while the ribs move up and out, so the chest space gets larger

17. Why must even a deeply sleeping person keep breathing, although their muscles are almost motionless?

- (A) cells stop respiring during sleep, so the body must store the night's air
- (B) every living cell goes on respiring to stay alive and needs a steady supply of oxygen, even at rest
- (C) the lungs use the night to make glucose and need air for it
- (D) breathing is only needed during movement, so sleep breathing is just a habit

18. A burning candle is lowered into a jar of air a student has breathed OUT and into a jar of fresh room air. The candle is snuffed out sooner in the exhaled-air jar. This is best explained by the fact that exhaled air has:

- (A) more oxygen, which made the candle burn itself out faster
- (B) less oxygen and more carbon dioxide than fresh air, so it supports burning less well
- (C) no oxygen at all, which is why a flame cannot survive in it
- (D) more water vapour, which is the substance that put out the flame

19. A cockroach is sealed in a jar. Air enters through tiny openings along its body and travels through fine tubes directly to its tissues. Naming first the openings and then the tubes, these are:

- (A) spiracles and tracheae
- (B) stomata and lenticels
- (C) gills and bronchi
- (D) nostrils and alveoli

20. Why must a fish keep moving water over its gills almost without stopping, while a human at the same level of activity breathes air far less often?

- (A) fish respire many times faster than humans, so they need oxygen more often
- (B) gills can only take oxygen from moving water and never from still water
- (C) water holds much less dissolved oxygen than air, so far more water must pass the gills to gather enough
- (D) water is heavier, so it carries the oxygen away before the gills can grab it

21. A frog can stay submerged in a pond for a long time without surfacing, while a fish moved onto dry land soon suffocates even though there is air around it. The reason for BOTH facts is that:

- (A) a frog has gills like a fish, and a fish has lungs that fail in air
- (B) both animals stop respiring when out of their usual surroundings
- (C) a frog can take in oxygen through its moist skin underwater, while a fish's gills can only extract oxygen dissolved in water, not from air
- (D) a frog stores oxygen in its stomach, while a fish has no need for oxygen on land

22. Why does a deep cut across the windpipe make breathing very hard, even when the lungs and diaphragm are perfectly healthy?

- (A) the windpipe is where energy is released, so the cells starve when it is cut
- (B) the windpipe makes the oxygen that the lungs themselves cannot make
- (C) the windpipe pumps the blood, so breathing fails when it is damaged
- (D) the windpipe is the main passage carrying air to the lungs, so damaging it blocks the air's route

23. A swimmer takes one deep breath and dives, then keeps her leg muscles working hard underwater. As the dive goes on, which kind of respiration becomes increasingly likely in those leg muscles?

- (A) photosynthesis, because the working muscles make their own oxygen
- (B) anaerobic respiration, because the oxygen runs short and lactic acid begins to form
- (C) aerobic respiration with no change, because one breath lasts the whole dive
- (D) no respiration at all, so that the body can save its oxygen

24. Match each organism to the structure it uses to take in oxygen. Which set is entirely correct?

- (A) earthworm — moist skin; cockroach — spiracles; fish — gills
- (B) earthworm — gills; cockroach — stomata; fish — moist skin
- (C) earthworm — lungs; cockroach — gills; fish — spiracles
- (D) earthworm — spiracles; cockroach — moist skin; fish — lenticels

25. A green leaf is sealed in a clear box in bright midday sunlight. Which is the MOST complete statement of what is happening to its gases?

- (A) it only photosynthesises in the daytime and does not respire at all
- (B) it only respire in the daytime and stops photosynthesising
- (C) it both respire and photosynthesises, but in strong light photosynthesis is greater, so overall it takes in carbon dioxide and gives out oxygen
- (D) it neither respire nor photosynthesises while it is sealed in the box

26. Air a person breathes OUT, compared with the air they breathe IN, has:

- (A) more oxygen, less carbon dioxide, and less water vapour
- (B) no oxygen at all and nothing but carbon dioxide
- (C) exactly the same amounts of every gas as the air breathed in
- (D) less oxygen, more carbon dioxide, and more water vapour

27. After a fermentation experiment, a lighted splint is held over the gas standing above the bubbling liquid and is snuffed out at once. The same gas also turns limewater milky. The gas is:

- (A) oxygen, which makes a glowing splint burst back into flame
- (B) hydrogen, which burns with a sharp popping sound
- (C) carbon dioxide, which does not support burning and turns limewater milky
- (D) nitrogen, which the yeast breathes out as it ferments

28. The same sealed clear box with a green leaf is now kept in complete darkness all night. By morning the air inside has:

- (A) more oxygen and less carbon dioxide, because the plant photosynthesises at night
- (B) more carbon dioxide and less oxygen, because in the dark only respiration occurs
- (C) no change in any gas, because plants do nothing while it is dark
- (D) more oxygen, because respiration stops once the light is removed

29. Put these inhalation events in correct cause-and-effect order: (i) air flows into the lungs; (ii) the chest space gets larger; (iii) the diaphragm flattens and the ribs rise; (iv) the pressure inside the lungs drops.

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|-----------------------|-----------------------|
| (A) iii → ii → iv → i | (B) i → iii → ii → iv |
| (C) ii → iv → i → iii | (D) iv → iii → ii → i |

30. Soaked germinating peas are sealed in a flask beside a small dish of limewater for a day; identical boiled (killed) peas are sealed in a matching flask as a control. The limewater milkens only in the live-pea flask. The control of boiled peas is included mainly to show that:

- (A) boiled peas release more carbon dioxide than live ones
- (B) limewater milkens on its own without any peas present
- (C) the carbon dioxide comes from the living peas respiring, not from the flask or the seeds simply sitting there
- (D) the live peas were photosynthesising in the sealed dark flask

31. At sea level a hiker breathes 16/min standing still; high on a mountain he breathes 24/min while still standing still. The best Class-7 explanation for the higher rate at altitude is:

- (A) the mountain air is full of carbon dioxide that must be cleared quickly
- (B) the cold air forces the diaphragm to move faster than usual
- (C) the lungs shrink in the cold and hold less air per breath
- (D) the air there is thinner, so each breath carries less oxygen and more breaths are needed to get enough

32. A respirometer holds a small living grasshopper in a sealed chamber. Over an hour a coloured drop in an attached tube moves steadily inward, showing a gas is being used up (the carbon dioxide given off is absorbed by a chemical). This steady use of gas is direct evidence that the grasshopper is:

- (A) photosynthesising, since it is giving out oxygen into the chamber
- (B) only breathing mechanically, with no respiration happening inside it
- (C) respiring continuously, taking in oxygen to release energy
- (D) storing the oxygen unused, which is why the drop moves

33. A sleeping baby keeps breathing steadily. A doctor explains it is the same fundamental reason a sealed jar of germinating seeds in the dark turns limewater milky. That shared reason is that:

- (A) all living cells, in animals and in plants alike, respire continuously and so use oxygen and release carbon dioxide
- (B) only animals respire, while seeds merely release stored gas
- (C) respiration only happens during movement, so both must be moving
- (D) plants and babies both photosynthesise at night to make energy

34. Which comparison of breathing structures across animals is CORRECT?

- (A) a cockroach uses gills while a fish uses spiracles
- (B) an earthworm uses lungs while a frog uses only gills
- (C) a fish uses gills while a cockroach uses tracheae fed through spiracles
- (D) a frog uses spiracles while a fish uses moist skin

- 35.** A glucose molecule is broken down in two different ways: with oxygen it yields carbon dioxide, water and energy; without oxygen in muscle it yields lactic acid and energy. The single most important difference for the body is that:
- (A) the oxygen-free route releases more energy because lactic acid stores energy
 - (B) the oxygen route releases much more usable energy because the glucose is broken down completely
 - (C) both routes release identical energy because they start from the same glucose
 - (D) the oxygen-free route is preferred at rest because it is cleaner
- 36.** The hair and sticky mucus that line the nasal cavity matter to respiration because they:
- (A) add extra oxygen to the air as it passes through the nose
 - (B) release energy from the incoming air for the body to use
 - (C) turn the carbon dioxide in the air back into oxygen
 - (D) trap dust and germs so that cleaner air reaches the lungs
- 37.** In which plant parts does gas exchange for respiration mainly take place — in the leaves, in the woody stem, and at the roots, respectively?
- (A) stomata; lenticels; air spaces between the soil particles
 - (B) lenticels; stomata; gills in the soil
 - (C) spiracles; stomata; alveoli
 - (D) stomata; alveoli; lenticels
- 38.** During a quiet, ordinary exhalation, which set correctly describes the ribs, the diaphragm, and the pressure inside the lungs?
- (A) ribs move up and out, the diaphragm flattens, and the pressure inside falls
 - (B) ribs stay still, the diaphragm flattens, and the pressure inside falls
 - (C) ribs move down and in, the diaphragm flattens, and the pressure inside falls
 - (D) ribs move down and in, the diaphragm domes up, and the pressure inside the lungs rises
- 39.** A burning candle, a respiring mouse, and germinating seeds are each sealed in identical jars beside limewater. After an hour the limewater in all three jars has turned milky. The single idea this demonstrates is that:
- (A) only the candle releases carbon dioxide, while the living things absorb it
 - (B) the seeds give out oxygen while the candle and mouse give out carbon dioxide
 - (C) burning and respiration in both the animal and the plant seeds all release carbon dioxide
 - (D) carbon dioxide is produced only when something is actually on fire
- 40.** A wind-instrument player blows out a long musical phrase, then gasps for breath afterwards. The gasping mainly serves to:
- (A) release stored alcohol that the lungs made during the long phrase
 - (B) cool the warm air inside the windpipe after the effort

(C) make fresh glucose to refill the tired chest muscles

(D) quickly take in oxygen and remove the carbon dioxide that built up while blowing

41. A green pond plant is sealed in a jar of water and left in bright sunlight; bubbles of gas collect at the top, and a glowing splint placed in the collected gas bursts back into flame.

The gas and the process producing the surplus of it are:

(A) carbon dioxide, from the plant respiring in the light

(B) oxygen, from the plant respiring faster than it photosynthesises

(C) oxygen, from photosynthesis outpacing respiration in the bright light

(D) carbon dioxide, from the plant fermenting the water like yeast

42. A boy breathes 18 times per minute at rest and 30 times per minute while cycling hard. By what fraction does his breathing rate increase from rest to cycling?

(A) by one-half of the resting rate

(B) by the whole resting rate (it doubles)

(C) by two-thirds of the resting rate

(D) by one-third of the resting rate

43. A girl holds her breath as long as she can, then is forced to breathe out hard. Which substance building up in her blood mainly triggers the overpowering urge to breathe?

(A) extra oxygen that her cells were unable to use

(B) lactic acid made directly by her lungs

(C) alcohol released from her resting muscles

(D) carbon dioxide produced by her respiring cells

44. An earthworm and a fish are both placed in waterlogged, oxygen-poor mud overnight. The earthworm crawls up to the surface while the fish, kept in the open water above, stays put. Which respiration-based explanation fits BOTH?

(A) the earthworm wants to drink rainwater while the fish dislikes the surface

(B) the earthworm's moist skin cannot get enough oxygen in air-starved soil so it surfaces, while the fish can still draw the little dissolved oxygen from the open water with its gills

(C) the earthworm switches to gills at the surface while the fish switches to skin

(D) both come up because the mud has too much oxygen to bear

45. A diagram shows a balloon-like air sac wrapped in fine blood vessels, where oxygen crosses from the air into the blood. The thin walls of this structure matter because they:

(A) are thick so they can store oxygen for a long time

(B) make the oxygen by breaking down the air inside them

(C) pump the blood around the body like the heart

(D) are thin enough to let oxygen pass easily and quickly into the blood

46. A student insists, 'Fish breathe water the way we breathe air.' The most accurate correction is:

- (A) fish breathe the water straight into their lungs like we breathe air
- (B) fish take in oxygen that is dissolved in the water, using their gills, rather than breathing the water itself
- (C) fish make their own oxygen out of the water as it passes the gills
- (D) fish do not need oxygen, so they simply pass water through for cooling

47. Both a fish in a river and a person on a hill ultimately need oxygen for the very same purpose. That shared purpose is to:

- (A) make their food by photosynthesis using the oxygen
- (B) produce glucose inside their breathing organs
- (C) release energy from food inside their cells by respiration
- (D) turn carbon dioxide directly into stored energy

48. Which observation would BEST convince a doubter that plants respire and do not merely photosynthesise?

- (A) a leaf in bright sunlight gives off bubbles of oxygen from a pond plant
- (B) a sealed jar of germinating seeds kept in the dark turns limewater milky and snuffs out a candle
- (C) a plant grows taller over several weeks in a sunny garden
- (D) a flower opens in the morning and closes again at night

49. A boy counts 20 breaths in 40 seconds while walking. Expressed as a breathing rate in breaths per minute, this is:

- (A) 20 breaths per minute
- (B) 30 breaths per minute
- (C) 13 breaths per minute
- (D) 60 breaths per minute

50. Why does the glucose in our food need the oxygen brought by breathing before most of its energy can be released?

- (A) oxygen first turns the glucose into a larger amount of glucose
- (B) oxygen is itself the energy, stored ready inside every cell
- (C) oxygen cools the cell so the glucose does not burn itself away
- (D) with oxygen the glucose can be broken down completely into carbon dioxide and water, which releases the most energy

51. A glass tube of clear limewater is connected so a person can both breathe in through it and breathe out through it (using a two-way mouthpiece). After many breaths the limewater turns milky. The fairest conclusion about why is that:

- (A) the inhaled air alone milkens it, proving room air is rich in carbon dioxide
- (B) the limewater milkens from the water vapour in the breath, not carbon dioxide
- (C) breathing in and out are identical, so the milkiness is purely chance
- (D) the exhaled air passing through adds much more carbon dioxide than the inhaled air removes, so carbon dioxide builds up and milkens the limewater

52. A trained athlete's resting breathing rate is 12/min, while an untrained person of the same age rests at 18/min. After identical light exercise both rise to 24/min. Comparing the rises, the athlete's breathing rate has increased by:

- (A) a larger fraction of its resting value than the untrained person's
- (B) a smaller fraction of its resting value than the untrained person's
- (C) exactly the same fraction as the untrained person's
- (D) no change, since both end at the same rate

53. A fish tank's water is gently heated. Warm water holds less dissolved oxygen than cool water. As the water warms, you would expect the fish to:

- (A) gulp and pump water over its gills faster, to gather enough oxygen from the oxygen-poorer warm water
- (B) slow its gill movements, because warm water carries oxygen more easily
- (C) stop using its gills and start breathing the air above the tank
- (D) need no oxygen at all once the water is warm

54. Which single statement about respiration is TRUE?

- (A) only animals respire, while plants merely photosynthesise
- (B) respiration stops in animals while they are asleep
- (C) in a human, only the cells of the lungs respire
- (D) every living cell of both plants and animals respire to release energy

55. A child says, 'A frog is an amphibian, so it must breathe with gills its whole life like a fish.' The best correction is:

- (A) an adult frog breathes only through gills, exactly like a fish
- (B) an adult frog breathes through its moist skin and its lungs, not gills
- (C) an adult frog breathes through spiracles like an insect
- (D) an adult frog does not breathe at all once it leaves the water

56. Which is the correct full word equation for aerobic respiration as taught in Class 7?

- (A) glucose + oxygen → carbon dioxide + water + energy
- (B) glucose → carbon dioxide + water + energy
- (C) glucose + carbon dioxide → oxygen + water + energy
- (D) carbon dioxide + water + energy → glucose + oxygen

57. A sealed jar holds a small burning candle; a second identical sealed jar holds a respiring grasshopper. In time the candle flame dies and the grasshopper becomes distressed. The common reason is that BOTH:

- (A) released oxygen and used up the carbon dioxide in the jar
- (B) needed the sunlight that the jar was blocking out
- (C) used up the oxygen and built up carbon dioxide in the sealed jar
- (D) kept producing glucose until the jar was completely full

58. Why does air actually rush INTO the lungs during inhalation, rather than the lungs 'sucking' it in by themselves?

- (A) the lungs grip the air like a magnet and pull it inward
- (B) the diaphragm pushes the air up into the lungs from the stomach
- (C) the chest expands so the pressure inside the lungs falls below the outside air pressure, and air flows from high to low pressure
- (D) warm air outside naturally floats up into the cooler lungs

59. Brewers use the same yeast that bakers use, but for a different end product: brewers want the alcohol while bakers want the dough to rise. The single process that serves both purposes is:

- (A) photosynthesis by the yeast, which makes sugar and oxygen
- (B) aerobic respiration of sugar, which makes only carbon dioxide and water
- (C) the boiling of the sugar liquid, which gives off alcohol vapour and steam
- (D) anaerobic respiration of sugar by the yeast, which yields both alcohol and carbon dioxide

60. While a person holds a slow, steady breath at the end of an inhalation, the diaphragm and rib muscles are still gently contracted to keep the chest enlarged. The moment they relax, what happens and why?

- (A) the chest enlarges further, lowering the pressure, so more air rushes in
- (B) the chest shrinks, raising the pressure inside the lungs above the outside, so air is pushed out — exhalation
- (C) nothing changes, because relaxed muscles keep the chest exactly as it was
- (D) the lungs make extra oxygen that forces the held air back out

Solutions — Science (Class 7), Chapter 10: Respiration in Organisms — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	A	21	C	31	D	41	C	51	D
2	D	12	A	22	D	32	C	42	C	52	A
3	A	13	A	23	B	33	A	43	D	53	A
4	A	14	D	24	A	34	C	44	B	54	D
5	B	15	A	25	C	35	B	45	D	55	B
6	B	16	D	26	D	36	D	46	B	56	A
7	B	17	B	27	C	37	A	47	C	57	C
8	C	18	B	28	B	38	D	48	B	58	C
9	C	19	A	29	A	39	C	49	B	59	D
10	B	20	C	30	C	40	D	50	D	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) Food such as glucose stores chemical energy in its bonds; respiration breaks the glucose down (with oxygen, aerobically) and releases that stored energy in a usable form. Energy is not created from nothing in cells, nor is it absent from food until oxygen reaches the lungs, and the stomach digests rather than destroys the energy.

2. (D) Aerobic breakdown (with oxygen) splits glucose fully into carbon dioxide and water and yields far more energy per glucose than anaerobic breakdown, which only partly breaks the glucose down into products such as alcohol or lactic acid. Equal starting glucose does not mean equal energy released, and the partial products do not store extra usable energy for the cell.

3. (A) Breathing is the mechanical movement of air in and out; respiration is the chemical release of energy inside cells. Faster breathing supplies extra oxygen so the muscle cells can respire faster, but the energy is released in those cells, not in the lungs. The two

processes are linked, not unconnected, and faster breathing supports rather than slows respiration.

4. (A) Air enters the nostrils, is cleaned and warmed in the nasal cavity, passes down the windpipe (trachea), branches into the bronchi, reaches the thin-walled alveoli, and there oxygen crosses into the blood. The nasal cavity comes before the trachea, and the trachea comes before the bronchi, so paths that reverse these are wrong.

5. (B) Respiring seeds use oxygen and give out carbon dioxide; the carbon dioxide turns limewater milky, directly evidencing respiration even without visible breathing. Feeling cool/dry, floating, or a candle merely staying lit are not tests for the carbon dioxide that respiration produces.

6. (B) During hard, fast effort the oxygen supply cannot keep up, so muscle cells partly respire anaerobically and accumulate lactic acid, causing cramp. Rest restores oxygen, which breaks the lactic acid down and relieves the cramp. Human muscles make lactic acid (not alcohol) anaerobically, and the cramp is caused by too little oxygen, not too much.

7. (B) Yeast in dough respire the sugar without enough oxygen (anaerobic respiration), giving off carbon dioxide; the trapped bubbles make the dough rise. Yeast does not photosynthesise (it is not green and has no chlorophyll), and the rise is from gas bubbles, not boiling water vapour.

8. (C) Lactic acid built up while oxygen ran short during exertion; continued fast breathing after stopping supplies the extra oxygen needed to break that lactic acid down. The lungs are not mechanically stuck, human muscles make lactic acid (not alcohol), and the goal is to clear lactic acid, not to make more.

9. (C) The values rise from 13 (asleep) to 40 (sprinting) as activity increases, showing breathing rate climbs with muscular effort to meet higher oxygen demand and clear more carbon dioxide. The data do not show the rate highest at rest, falling with activity, or fastest during sleep.

10. (B) The rubber sheet represents the diaphragm. Pushing it up models the diaphragm doming up (relaxing), which shrinks the chest space and forces air out of the balloon — exhalation. Pulling the sheet down would model inhalation; pushing up does not enlarge the chest space, and this models breathing, not the heartbeat.

11. (A) Limewater turning milky indicates carbon dioxide. Y turning slightly milky shows ordinary air contains a little carbon dioxide, while X turning milky much faster shows exhaled air contains far more. So room air is not free of carbon dioxide, it does not contain more than exhaled air, and the change is due to carbon dioxide, not water vapour.

12. (A) Oxygen falls from about 21% to 16% (some used, but most still present) while carbon dioxide rises from about 0.04% to 4% (a large increase). So exhaled air is not

oxygen-free, the body does not add oxygen, and the amounts clearly change — evidence that respiration consumed oxygen and produced carbon dioxide.

13. (A) A trained body supplies and uses oxygen more efficiently, so it meets the same oxygen demand with fewer breaths — A's lower rate reflects fitness. A higher rate does not prove stronger lungs; respiration has not stopped in A; and all working bodies, trained or not, still need oxygen.

14. (D) Anaerobic respiration (fermentation) by yeast yields alcohol (the smell) and carbon dioxide (the fizz and the inflated balloon) — one process explains all three. Yeast does not photosynthesise, the changes are chemical not boiling, and the yeast is breaking sugar down, not rebuilding it into sugar.

15. (A) 15 breaths per minute times 45 minutes equals 675 breaths. Adding 15 and 45 gives 60; multiplying by an extra factor of ten gives 6750; and 450 corresponds to only 30 minutes, not 45.

16. (D) On inhaling, the diaphragm contracts and flattens downward and the ribs swing up and out, enlarging the chest space so air is drawn in. Doming up and ribs moving down would shrink the space (exhalation), the diaphragm cannot flatten and let ribs move in at once, and the lungs do not move air by themselves.

17. (B) Respiration in cells never stops in a living body; even at rest the cells need oxygen and release carbon dioxide, so breathing must continue. Cells do not stop respiring in sleep, lungs do not make glucose, and breathing is essential at all times, not only during movement.

18. (B) Burning needs oxygen. Exhaled air has less oxygen (used in respiration) and more carbon dioxide, so the candle dies sooner. Exhaled air does not have more oxygen, it is not oxygen-free (the candle does burn briefly), and it is the lower oxygen, not water vapour, that snuffs the flame.

19. (A) Insects breathe through small openings called spiracles, and air passes through a branching network of tubes called tracheae straight to the tissues. Stomata and lenticels are plant pores; gills, bronchi, nostrils and alveoli belong to other animals' systems, not the insect tracheal system.

20. (C) A given volume of water contains far less oxygen than the same volume of air, so a fish must pass a large amount of water over its gills to extract enough. It is not that fish respire vastly faster, nor that gills physically cannot use still water, nor that water's weight sweeps oxygen away.

21. (C) A frog's moist skin lets it absorb dissolved oxygen underwater, so it need not surface; a fish's gills work only on oxygen dissolved in water and collapse in air, so the fish suffocates on land despite the surrounding air. Frogs do not have gills as adults storing

oxygen in the stomach, and both animals keep respiring; their organs simply suit different surroundings.

22. (D) The trachea (windpipe) is the chief airway between the throat and the lungs; if it is blocked or cut, air cannot reach healthy lungs and breathing fails. Energy is released in cells (not the windpipe), the windpipe does not manufacture oxygen, and it carries air, not blood.

23. (B) As the single breath's oxygen is used up during sustained effort, the muscles increasingly respire anaerobically, producing lactic acid. Muscles cannot photosynthesise, one breath cannot keep full aerobic respiration going indefinitely, and living muscle cannot simply stop respiring.

24. (A) An earthworm absorbs oxygen through its moist skin, a cockroach takes air through spiracles, and a fish uses gills. The other sets misassign these — earthworms lack gills/lungs/spiracles, cockroaches do not use stomata or moist skin, and fish use neither spiracles nor lenticels.

25. (C) A leaf in bright light does both: it respire (always) and photosynthesises (in light). Because photosynthesis outpaces respiration in strong light, the net effect is carbon dioxide taken in and oxygen given out. The plant does not stop respiring by day, does not stop photosynthesising in light, and being sealed does not switch either process off.

26. (D) Respiration uses oxygen and produces carbon dioxide and water, so exhaled air has less oxygen, more carbon dioxide, and more water vapour than inhaled air. It is not richer in oxygen, not oxygen-free (it still holds about 16% oxygen), and clearly differs from inhaled air.

27. (C) A gas that snuffs a flame and turns limewater milky is carbon dioxide. Oxygen would relight a glowing splint, hydrogen pops when lit, and yeast does not 'breathe out' nitrogen — fermentation gives off carbon dioxide.

28. (B) Without light a plant cannot photosynthesise, but it goes on respiring all night — using oxygen and releasing carbon dioxide. So the air gains carbon dioxide and loses oxygen. The plant does not photosynthesise in the dark, it does not become inactive, and respiration continues (it does not stop) in darkness.

29. (A) First the muscles act (diaphragm flattens, ribs rise), enlarging the chest space, which lowers the pressure inside the lungs, and only then air rushes in: iii → ii → iv → i. Air cannot flow in first, the chest cannot enlarge before the muscles move it, and the pressure does not drop before the chest expands.

30. (C) A control with dead (boiled) peas isolates the variable: since only the live peas milken the limewater, the carbon dioxide must come from living respiration, not the apparatus or mere presence of seeds. Boiled peas do not respire (so release no carbon

dioxide), limewater does not milken by itself, and sealed peas in the dark are respiring, not photosynthesising.

31. (D) High-altitude air is thinner (less dense), so each breath delivers less oxygen, and the body breathes more often to make up the shortfall. Mountain air is not rich in carbon dioxide, cold does not by itself drive the diaphragm faster, and lungs do not physically shrink in cold to cause this.

32. (C) The drop moves inward because oxygen is being consumed (the carbon dioxide is mopped up), showing the animal steadily uses oxygen — the hallmark of ongoing respiration to release energy. An animal does not photosynthesise, breathing without respiration would not consume oxygen, and the oxygen is being used, not merely stored.

33. (A) Both cases show the universal feature of life: every living cell respire all the time, using oxygen and giving off carbon dioxide — hence the baby's steady breathing and the seeds milkening limewater. Plants respire too (not only animals), respiration is not limited to movement, and neither baby nor seeds photosynthesise in the dark.

34. (C) A fish breathes with gills; a cockroach draws air through spiracles into a network of tracheae. Cockroaches do not use gills nor fish spiracles, earthworms have no lungs and adult frogs no gills, and frogs do not use spiracles nor fish moist skin.

35. (B) Complete breakdown with oxygen (to carbon dioxide and water) releases far more energy per glucose than the incomplete anaerobic route ending in lactic acid. Lactic acid is a leftover, not a richer energy store; equal glucose does not mean equal energy; and at rest the body favours the high-yield aerobic route, not the anaerobic one.

36. (D) Nasal hairs and mucus filter dust and germs from incoming air, protecting the lungs by delivering cleaner air. The nose does not add oxygen, release energy (that happens in cells), or convert carbon dioxide into oxygen.

37. (A) Leaves exchange gases through stomata, woody stems through lenticels, and roots take in oxygen from air held in the spaces between soil particles. Gills, spiracles and alveoli are animal structures, and the leaf/stem pores must not be swapped.

38. (D) On exhaling, the ribs drop down and in and the diaphragm relaxes upward (domes up), shrinking the chest so the pressure inside the lungs rises and air is pushed out. Ribs moving up/out and a flattening diaphragm describe inhalation, and a flattening diaphragm cannot accompany shrinking and rising pressure.

39. (C) All three milken the limewater, so all three release carbon dioxide — the candle by burning, the mouse and seeds by respiration. The living things do not absorb the carbon dioxide here, the seeds release it (not oxygen) in the dark, and carbon dioxide comes from respiration too, not only from fire.

40. (D) Holding a long phrase delays normal breathing, so oxygen runs low and carbon dioxide builds up; the gasping restores oxygen and clears the carbon dioxide. The lungs do not store alcohol, the purpose is gas exchange not cooling, and breathing does not manufacture glucose.

41. (C) A glowing splint relighting identifies oxygen. In bright light photosynthesis exceeds respiration, so the plant gives off surplus oxygen. Carbon dioxide would not relight a splint, respiration consumes rather than releases oxygen, and aquatic plants do not ferment water.

42. (C) The increase is $30 - 18 = 12$ breaths; as a fraction of the resting 18 that is $12/18 = 2/3$. One-half would be 9, doubling would be a rise of 18, and one-third would be 6 — none match the actual rise of 12.

43. (D) While the breath is held, cells keep respiring and carbon dioxide builds up in the blood, triggering the urge to breathe. Oxygen is being used up, not piling up; lungs do not make lactic acid; and resting human muscles do not release alcohol.

44. (B) Waterlogged mud has lost the air spaces, so the worm's skin cannot collect enough oxygen and it surfaces; the fish, in open water, can still extract dissolved oxygen through its gills. Worms surface for oxygen, not to drink; worms have no gills and fish no skin breathing; and the problem is too little oxygen, not too much.

45. (D) These are the alveoli; their very thin walls let oxygen diffuse easily and quickly into the surrounding blood. Thin walls do not store gas, alveoli do not manufacture oxygen, and they do not pump blood (that is the heart's job).

46. (B) Gills extract oxygen already dissolved in the water; the fish does not 'breathe' water itself. Fish have no lungs taking in water, they cannot make oxygen from water, and they do need oxygen — water flow is for gas exchange, not cooling.

47. (C) Whatever the organ — gills or lungs — the oxygen is ultimately used by cells to release energy from food (glucose) through respiration. Animals do not photosynthesise, breathing organs do not make glucose, and carbon dioxide is a waste product, not turned into energy.

48. (B) Seeds in the dark cannot photosynthesise, so the carbon dioxide that milkens limewater and the oxygen loss that snuffs the candle can only come from respiration — clinching proof that plants respire. Oxygen bubbles in light show photosynthesis, growth and flower movements are not direct tests for respiration.

49. (B) 20 breaths in 40 seconds is $20 \div 40 = 0.5$ breaths per second; over 60 seconds that is 30 breaths per minute. Taking 20 ignores the time scaling, 13 corresponds to scaling the wrong way, and 60 would need 20 breaths in only 20 seconds.

50. (D) Oxygen allows complete breakdown of glucose to carbon dioxide and water, the route that frees the most energy; without it only partial breakdown (and less energy) is possible. Oxygen does not multiply glucose, is not itself energy, and its role is chemical breakdown, not cooling.

51. (D) Exhaled air is far richer in carbon dioxide than inhaled air, so over many breaths net carbon dioxide accumulates and turns the limewater milky. Inhaled (room) air alone has only a trace of carbon dioxide, the milkiness is due to carbon dioxide not water vapour, and inhaled and exhaled air differ markedly — the result is not chance.

52. (A) The athlete rises 12→24, an increase of 12 on a base of 12 (a 100% rise); the untrained rises 18→24, an increase of 6 on a base of 18 (about a 33% rise). The athlete's fractional increase is larger, not smaller or equal, and both clearly do change.

53. (A) Less dissolved oxygen means the fish must move more water over its gills to extract enough, so gill pumping speeds up. Warm water does not carry oxygen more easily, a fish cannot switch to breathing air, and it still needs oxygen for respiration.

54. (D) Respiration to release energy occurs in every living cell of plants and animals alike. Plants respire too (not only photosynthesise), respiration does not stop during sleep, and all of a human's cells respire, not just the lung cells.

55. (B) An adult frog takes in oxygen through its moist, moist skin and through its lungs — it does not keep fish-like gills. It does not use insect spiracles, and it certainly continues to respire and breathe out of water.

56. (A) Aerobic respiration combines glucose with oxygen to give carbon dioxide, water and energy. Leaving out oxygen describes the anaerobic case, swapping oxygen for carbon dioxide on the left is wrong, and reversing the arrow describes photosynthesis, not respiration.

57. (C) Both the flame and the animal consume oxygen and give off carbon dioxide; in a sealed jar the oxygen runs out and carbon dioxide accumulates, so the flame dies and the animal struggles. Neither releases oxygen or absorbs carbon dioxide, neither needs sunlight here, and neither makes glucose.

58. (C) Enlarging the chest lowers the pressure inside the lungs below outside pressure, and air flows in from high to low pressure — the lungs do not actively pull air. Lungs are not magnets, the diaphragm enlarges the chest rather than pushing air from the stomach, and air flow here is driven by pressure, not temperature.

59. (D) Yeast respiring sugar without enough oxygen produces both alcohol (valued by brewers) and carbon dioxide (which raises bakers' dough), so one anaerobic process serves both. Yeast does not photosynthesise, aerobic respiration would not make the alcohol, and the products come from respiration, not boiling.

60. (B) Relaxing the diaphragm and rib muscles lets the chest spring back smaller, which raises the pressure inside the lungs above the outside and pushes air out — exhalation. Relaxing cannot enlarge the chest or keep it fixed, and the lungs do not manufacture oxygen to expel air.